

```
In [1]: import cv2
        from matplotlib import pyplot as plt
        import numpy as np
        import imutils
        import easyocr
```

1. Read in Image, Grayscale and Blur

```
In [2]: img = cv2.imread('/kaggle/input/car-plate-
        detection/images/Cars115.png')
        gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
        plt.imshow(cv2.cvtColor(gray, cv2.COLOR_BGR2RGB))
```

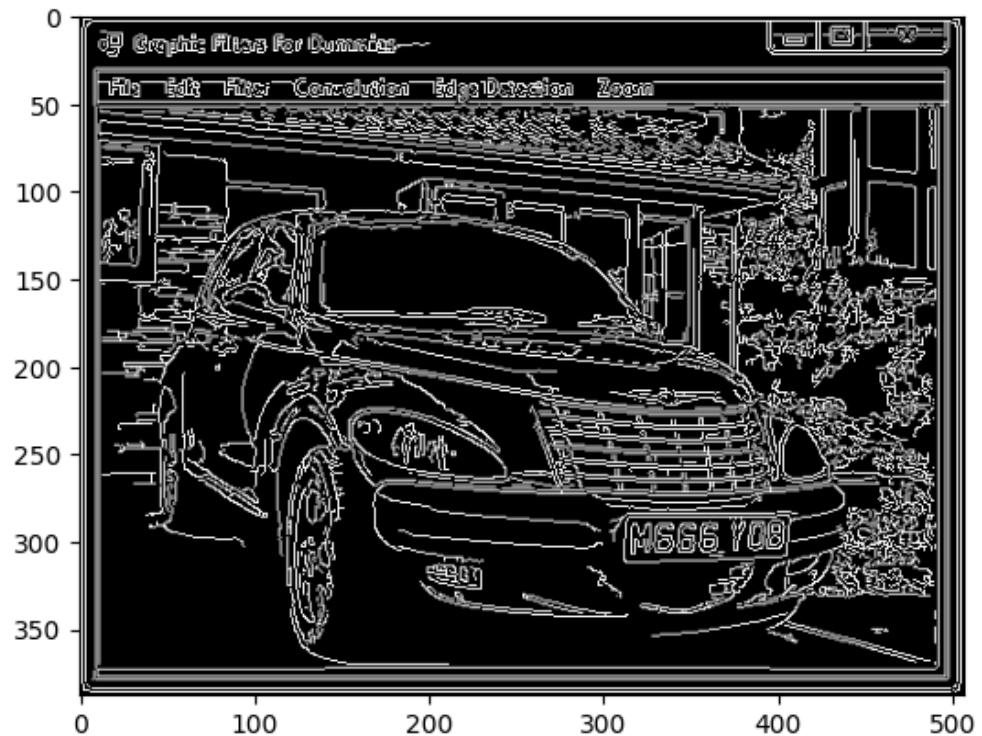
Out[2]: <matplotlib.image.AxesImage at 0x7fe254bb0650>



2. Apply filter and find edges for localization

```
In [3]: bfilter = cv2.bilateralFilter(gray, 11, 17, 17) #Noise reduction
        edged = cv2.Canny(bfilter, 30, 200) #Edge detection
        plt.imshow(cv2.cvtColor(edged, cv2.COLOR_BGR2RGB))
```

Out[3]: <matplotlib.image.AxesImage at 0x7fe247a004d0>



3. Find Contours and Apply Mask

```
In [4]: keypoints = cv2.findContours(edged.copy(), cv2.RETR_TREE,
        cv2.CHAIN_APPROX_SIMPLE)
        contours = imutils.grab_contours(keypoints)
        contours = sorted(contours, key=cv2.contourArea, reverse=True)[:10]
```

```
In [5]: location = None
        for contour in contours:
            approx = cv2.approxPolyDP(contour, 10, True)
            if len(approx) == 4:
                location = approx
                break
```

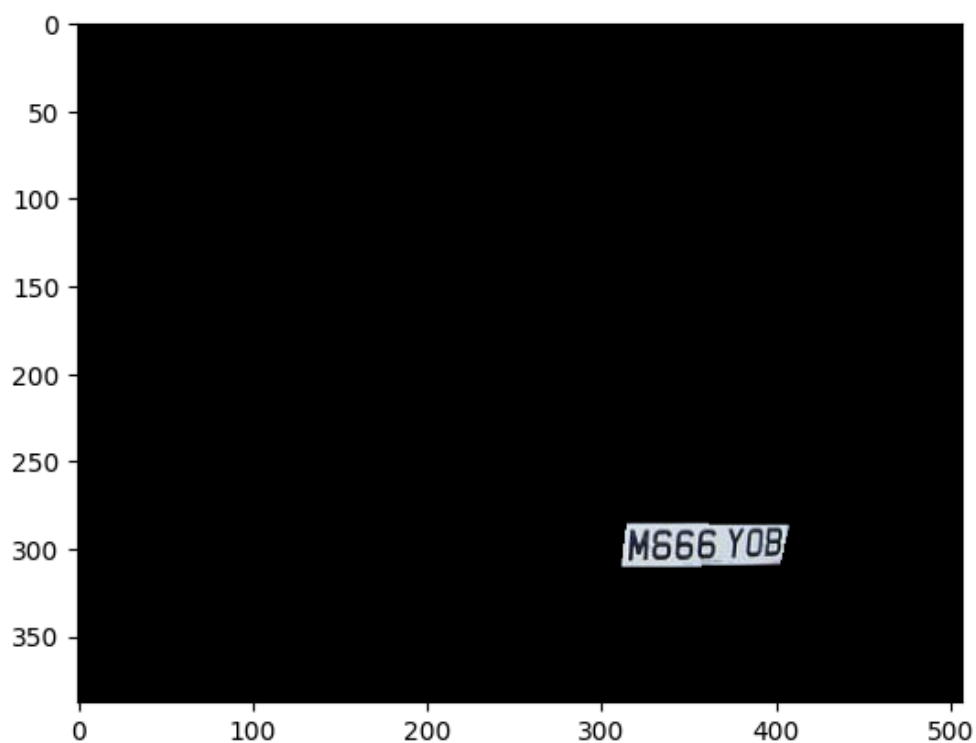
```
In [6]: location
```

```
Out[6]: array([[[[407, 287]],  
               [[402, 309]],  
               [[312, 310]],  
               [[315, 286]]], dtype=int32)
```

```
In [7]: mask = np.zeros(gray.shape, np.uint8)  
new_image = cv2.drawContours(mask, [location], 0,255, -1)  
new_image = cv2.bitwise_and(img, img, mask=mask)
```

```
In [8]: plt.imshow(cv2.cvtColor(new_image, cv2.COLOR_BGR2RGB))
```

```
Out[8]: <matplotlib.image.AxesImage at 0x7fe2476edb10>
```

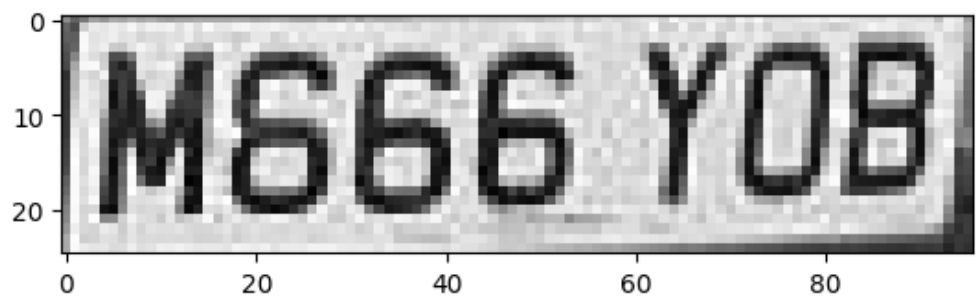


```
In [9]: (x,y) = np.where(mask==255)  
(x1, y1) = (np.min(x), np.min(y))  
(x2, y2) = (np.max(x), np.max(y))  
cropped_image = gray[x1:x2+1, y1:y2+1]
```

```
In      plt.imshow(cv2.cvtColor(cropped_image, cv2.COLOR_BGR2RGB))
```

```
[10]:
```

```
Out[10]: <matplotlib.image.AxesImage at 0x7fe247d55910>
```



4. Use Easy OCR To Read Text

```
In      reader = easyocr.Reader(['en'])  
[11]:   result = reader.readtext(cropped_image)  
       result
```

```
Out[11]: [([[0, 0], [96, 0], [96, 25], [0, 25]], 'Ms66 YOB',  
          0.8152418824567617)]
```

5. Render Result

```
In [12]: text = result[0][-2]
font = cv2.FONT_HERSHEY_SIMPLEX
res = cv2.putText(img, text=text, org=(approx[0][0][0], approx[1][0][1]+60), fontFace=font, fontScale=1, color=(0,255,0), thickness=2,
lineType=cv2.LINE_AA)
res = cv2.rectangle(img, tuple(approx[0][0]), tuple(approx[2][0]),
(0,255,0),3)
plt.imshow(cv2.cvtColor(res, cv2.COLOR_BGR2RGB))
```

Out[12]: <matplotlib.image.AxesImage at 0x7fe247d484d0>



In []:

In []: